

Fragmentation and firm choice: The effects of municipal decentralization on firm-level agglomeration economies*

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1 Introduction

Metropolitan regions in the United States are characterized by a distinct lack of centralized authority, composed instead of a patchwork of fragmented municipal governments. Indeed, the US Census of Governments records 35,673 different general purpose governments categorized as either municipalities or townships, a number that contrasts dramatically with the only 935 Census-recognized core-based statistical areas (CBSAs) that denote distinct city-regions. At the same time, not all regions are created equally fragmented: within the Miami metropolitan statistical area there exist 104 separate municipal entities, whereas the upstate region of Jacksonville counts only 16. Naturally, with such a high degree of variance in the political makeup of cities, a significant amount of energy has been devoted to identifying how the degree of municipal fragmentation affects various economic outcomes.

Fragmentation takes many forms. Apart from the proliferation of “general-purpose” governments, American metros also experience fragmented distributions of authority through special-purpose governments such as school districts (Cynamon & Pavel, 2022) and special districts (Foster, 1997) that overlap municipal borders. Often referred to as “vertical

*This is a partial draft of a senior honors thesis in economics, including the introduction, literature review, and a discussion of methods. Preliminary results are presented for first stage estimates, and all accompanying code can be accessed at <https://github.com/finnmeffe/honors-thesis>

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fragmentation”, this type of jurisdictional power-sharing involves multiple tiers of government operating in the same space. Conversely, the splintering of a metropolitan area along the boundaries of different municipal governments within the *same* tier is referred to as “horizontal” fragmentation. While the complex webs of responsibility that vertical fragmentation creates have found widespread critique in both the theoretical and empirical literature (Grassmueck & Shields, 2010; Goodman, 2019), more controversy has arisen on the subject of horizontal fragmentation. In particular, the economics literature offers many competing perspectives on the effects of fragmentation on economic welfare.

Such analyses concentrate themselves upon the implications of mobile households and the public goods provided to attract them within a competitive ecosystem for residents (“*voting with their feet*”). Complementing this, I choose to look at how municipalities interact with firms in metropolitan systems of inter-local competition. American municipalities generally have a robust set of tools available to incentivize firms to relocate, a fact brought into the national attention by the intense campaigns (or bidding wars) waged by local governments in 2017 to attract Amazon’s “HQ2”¹. While less attention-grabbing, a similar process plays out within metropolitan borders (Schragger, 2016), with competition for firms more often than not acquiring a local, rather than national, flavor.

Municipal fragmentation has been hypothesized to influence a variety of outcomes, from urbanization (Cappelli et al., 2021) to local spending (R. M. Hendrick et al., 2011) to long-run economic growth (Grassmueck & Shields, 2010). I focus on the spatial agglomeration of firms: does the jurisdictional “tapestry” change how firms cluster? I connect this to a lengthy literature suggesting strong agglomeration economies in which firms receive significant benefits from locating close to other firms, through mechanisms such as knowledge spillovers and labor pooling. Specifically, I direct my analysis in light of recent work on the *attenuation* of these benefits that suggests the effects of these agglomeration economies may disappear at

¹Over 200 cities submitted bids for the company’s new headquarters, with the 20 semi-finalists offering an average bid of \$2.15 billion in subsidies from cities and \$6.75 billion from states (Farren & Philpot, 2018) Amazon eventually chose a split location between Arlington, VA, and New York City, the latter of which was withdrawn after strong local opposition.

scales as small as thousands of meters. As such, even slight spatial rearrangements of firm location in response to municipal incentives may have dramatic effects on economic growth.

In this paper, I investigate the role of spatial agglomeration as a specific mechanism by which fragmentation affects economic growth. I first summarize the two major strains of literature - municipal public choice theory and agglomeration economies - I bridge across in this study. Next, I introduce a basic model of firm location and demonstrate that intra-local competition has the potential to upset the “natural” equilibrium of firm choice. In the empirical section, I provide an estimate of the impact of municipal fragmentation on firm agglomeration using an instrumental variables strategy similar to the “small-streams” approach developed by Hosxby (2000) and later used as an instrument for municipal boundaries by Hatfield et al. (2024), using small waterways as an exogenous determinant of jurisdictional borders. To gain a more accurate picture of jurisdictional fragmentation, I construct a novel spatial index using municipal borders alongside other, more common measures found in the literature. To assess spatial agglomeration, I match firms from County Business Patterns (CBP) data to census block level employment data, creating a probability-weighted series of discrete firm locations. I generate a set of global agglomeration measures at the CBSA level using Ripley’s K function and the m function to quantify firm clustering. I find negative, though insignificant effects of municipal fragmentation on the degree of firm clustering.

2 Literature Review

The contemporary economics literature offers a litany of interpretations of the welfare implications of municipal fragmentation, though two competing camps have emerged over the past decades (Agrawal et al., 2022). The first broadly includes those supporting a Tiebout-style hypothesis that competition between localities is capable of producing an efficient provision of public goods. Tiebout (1956) argued that household relocation - “voting with their feet” - drove a market for local governments that resulted in an equilibrium responsive to local

needs. A singular, central municipality, in contrast, is unable to tailor this package to any set of differentiated preferences among people, reducing potential welfare. Moreover, Brennan and Buchanan (1978) stressed the importance of an array of local governments in tempering the potential for fiscal exploitation by a central authority. On the other hand, Oates (2011) noted the potential for inefficient policy outcomes to arise from externalities between jurisdictions, complemented by Zodrow and Mieszkowski (1986), who viewed competition as a race to the bottom that dampens the ability of local governments to provide public goods.

Recent empirical studies measuring orthodox economic outcomes tend to support the virtues of intra-local competition, with Grassmuck and Shields (2010) using panel data to identify greater economic growth in more fragmented metropolitan areas; Hatfield et al. (2024) finds similar positive effects for housing values. At the same time, Goodman (2019) and Freemark et al. (2020) both tie fragmentation to inequality, complicating the macro-level economic picture. A major barrier to studying the influence of fragmentation on outcomes at the metropolitan level is the rigidity of municipal boundaries over the last 50 years. *Vertical* fragmentation has experienced large shifts, including the mass consolidation of school districts in an attempt to stave off rural depopulation during the 1950s-1970s and movements to create wide ranging “special districts,” such as water, housing, or fire authorities. Despite the changes to the web of overlapping authority, the actual *horizontal* dimension of municipal borders has remained stagnant Goodman (2019).

Following the theoretical basis laid out by Tiebout, much of the public choice literature has centered on the implications of household mobility, investigating the degree to which these responses of municipalities to residents create disparate outcomes through the mechanism of intergovernmental competition. However, relatively little attention has been devoted to the role that firms play in the municipal market. The 20th century equilibrium characterized by central business district (CBD) firm location and suburban households is rapidly changing in the contemporary American metropolis, with a growing share of suburban municipalities entering the marketplace for firms (Muller, 1997). Major firms are increasingly

headquartered outside of the CBD, in part due to the growing power of suburban municipalities to offer competitive tax bundles as seen in Schragger (2016).

Municipalities have a range of levers to pull in attempting to lure firms within their boundaries. Among these, tax breaks (both business and property) and service provision are the most common, and can be targeted to favor both particular firms or industries, or “economic development” in general Bartik (2017). The theory behind municipal competition for firms is well overviewed by Glaeser (2001), who articulates separate conditions under which fragmentation may both increase and decrease allocative efficiency. An important mechanism is the relative information and incentives of government and private actors: if municipalities are able to attract firms where their marginal effect is best felt and are incentivized purely by community benefit, then one would expect fragmentation to improve welfare. On the other hand, if benefits are not properly internalized, then intra-metropolitan competition could adversely impact welfare. This approach is expanded by Ferrari and Ossa (2023), who locate the “subsidy” framework within a quantitative economic geography model at the state level. Interstate subsidy effects are also investigated by Slattery and Zidar (2020) and Slattery (2025), both finding that subsidies may not generate sufficient welfare benefits to justify their existence as “place-based policy.” Mast (2020) complements this in a local context, finding that while municipal incentives do have some power to influence nearby firm location, the effects are highly local, and insufficient variance in policy choices by municipalities diminishes the strength of competition.

While growing competition between jurisdictions within metropolitan areas may be *prima facie* beneficial to the individual firm as it navigates towards lower tax burdens, this state of affairs is worrying in light of the large externalities or “spillover effects” generated by firms that they do not themselves internalize. Scholars as far back as have emphasized the role that industrial agglomeration plays in economic growth, with firm “clustering” often providing a conduit for growth. While there are numerous explanations for the mechanisms behind these externalities, the most intriguing is the role of knowledge spillovers: the theory that

flows of ideas between similarly situated firms are a catalyst for innovation and productivity. Glaeser et al. (1992) famously investigated the role of firm agglomeration in a nationwide analysis that affirmed its productive power, a result confirmed by a host of subsequent studies (Glaeser & Gottlieb, 2009; Ellison et al., 2010).

More recently, research has shed light on the explicit spatial range of agglomeration: at what point do the spillover effects accessed by proximity begin to attenuate? Rosenthal and Strange (2003) demonstrate strong local effects in which the benefits of firm entry diminish drastically past a five mile radius, whereas a host of more recent papers argue that this radius may be even smaller and limited to the range of 250-500 meters (Arzaghi & Henderson, 2008; Ahlfeldt et al., 2015; Coll-Martínez, 2019). Thus, while metro-level perspectives on agglomeration have little to say about jurisdictional fracturing within its borders, a more fine-grained analysis of firm spatial choice presents significant implications for the welfare effects of municipal fragmentation. Municipal subsidies could offer incentives for firms to relocate when they otherwise would have selected a location in greater proximity to similar firms in hopes of accessing spillovers. On the other hand, such incentives could create a coordinating strategy for similar firms to locate in municipalities that offer targeted subsidies, resulting in a degree of agglomeration that would not have been achieved due to a litany of other location choice factors.

3 Data

3.1 Fragmentation Measures

I define fragmentation loosely as the dispersion of political “power” among different municipalities in a metropolitan area. Interpretations of what this practically means are varied, and I will use a set of different measures to ensure I am capturing multiple facets of municipal authority. The most basic measure is simply the number of municipal governments per capita within a metropolitan area, which I calculate using the 2022 *Census of Gov-*

Table 1: Fragmentation Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Num. Counties	918	1.95	2.21	1	29
Num. Municipalities	918	14.79	28.29	0	376
Num. all Local Govs.	918	25.73	46.56	1	547
Municipal Pop.	918	220,816.00	834,089.00	0	13,587,249
Total Pop.	918	282,444.00	1,101,818.00	3,820	20,686,896
HHI - Municipal Pop.	918	0.50	0.25	0.00	1.00
HHI - Total Pop.	918	0.40	0.28	0.02	1.00
Muns. per Person	916	0.0002	0.0002	0.0000	0.001
Share in Central City	918	0.54	0.27	0.05	1.00

ernments data. This database catalogs each local government in the US. The *Census of Governments* data distinguishes between municipal and township governments at the sub-county general purpose level. While the set of municipal governments corresponds generally to Census Bureau incorporated places, definitions of what constitutes a township can vary widely between states in the scope of services provided (U.S. Census Bureau, n.d.). Thus, I focus primarily on fragmentation across municipalities, though I also gather total measures of both municipalities and townships for completeness. *Census of Governments* data also captures municipal populations and tax accounts, which are used to create more sophisticated measures of the distribution of people and finance across a given metropolitan area. I use a Herfindahl-Hirschmann Index (HHI), given by the equation:

$$HHI = \sum_{i=1}^n (x_i)^2$$

Where i is one of n municipalities in a CBSA, and x is the relative share of the variable of interest. For instance, if a metropolitan area has a single municipality with all of the population (such as Honolulu, HI), the HHI will be 1. Conversely, if the population is divided among 10 municipalities of equal size, the HHI will be 0.1.

CBSAs exhibit a high degree of heterogeneity, ranging from completely consolidated to

extremely fragmented. Basic summary statistics are given in Table 1. The average CBSA is moderately fragmented with 14.79 municipalities an average HHI of 0.50. Fragmentation increases, as expected, with the introduction of local governments into the data. Notably, while counting both municipalities and townships increases the average local governments by roughly 9, there is significantly smaller of a corresponding increase in total populations, demonstrating the generally smaller population sizes of townships. Additionally, the average CBSA has 0.54% of residents living within the central city, defined here as the largest city within the CBSA. Recognizing that these statistics are not population weighted, it cannot be inferred that these are the conditions faced within a typical city, but they do convey the fact that metropolitan areas face distinct environments within which competition for firms operates.

3.2 Spatial Difference Index

The statistics presented above reflect those most commonly found in empirical research on the effects of horizontal fragmentation; others include indices that capture the dispersion of fiscal power and granular metrics of population sorting (R. Hendrick & Shi, 2015). While these measures are able to capture telling information about metropolitan political ecosystems, some nuance is inevitably lost in the process. In particular, I add to measures of municipal *number* and relative size an understanding of shape, assembling a proxy for *how* municipal competition practically plays out.

The most basic intuition behind this process is that it aims to capture the “strangeness” of municipal boundaries: under a stylized model in which municipalities do not compete with each other but instead grow in a quasi-natural fashion (e.g. in the Alonso-Muth-Mills monocentric city model with a circular boundary), borders follow simple geometric bounds². Empirically, municipal boundaries in the United States bear little resemblance to these simple geometric patterns. The strange shapes of American municipalities have been noted

²With the pervasive organization of streets into a grid throughout the United States, these “ideal” municipalities could also be taken as rectangular (Alexander, 2024).

in popular publications such as The Economist (2018) - noting the particularly egregious example of Birmingham, Alabama (a complete metropolitan image is available in Figure 3 of Appendix B). Still, the role of borders has been relatively understated in academic literature: exceptions include Austin (1999) and Lichter et al. (2007). Lewis (1996) is perhaps the best example, describing the key role of urban political structures in mediating economic outcomes and sharply critiquing the fraught processes by which land is distributed within metropolitan regions. These boundaries reflect a convoluted legal and political history: municipalities adjust their territory through multiple mechanisms, with the most common being incorporation, annexation, and mergers (Haider-Markel, 2016). Moreover, state laws and idiosyncratic political histories can have deep effects on the construction of borders.

By quantifying the degree of “strangeness” in borders, I capture a snapshot of the competitive environment that municipalities exist within, with a more *messily* fragmented metropolitan area conveying information about the degree of competition that local governments face for territory and households. Municipal borders that diverge into geometric irregularity are those where fraught municipal expansions and contractions take place, such as the pockets created by “underbounding” analyzed by Lichter et al. (2007). I construct a spatial difference index to quantify this phenomenon. I calculate a municipality’s departure from an ideal shape, performing separate constructions for a minimum bounding circle and rectangle. Sample shapes for Birmingham can be found in Appendix B Figure 4. The index is given as follows:

$$\theta_j = \sum_i (1 - \lambda_i) \frac{A_i}{\sum_i A_i}, \quad \lambda_i = \frac{A_i}{A'_i}$$

Where θ_j gives the spatial difference index of CBSA j , which sums area A -weighted spatial differences λ_i for each municipality i , themselves composed of the percentage of “ideal” area A'_i occupied by the actual municipal boundaries A_i . I also calculate a simpler central cities variant, which only calculates the set of i ’s that constitute the central municipalities of a CBSA. I take these from the Census-designated name for the CBSA; for example, the

“Seattle–Tacoma–Bellevue, WA metropolitan statistical area” would have 3 central cities of Seattle, Tacoma, and Bellevue.

Table 2: Top and Bottom 10 MSAs by Spatial Fragmentation (Bounding Rectangle)

CBSA Title	SDI	SDI - Central	Num. Munis	Muni Pop.	HHI - Muni Pop.
Top 10					
<i>Wichita, KS</i>	0.66655	0.27873	50	577787	0.47521
<i>Tulsa, OK</i>	0.66321	0.43184	73	807991	0.28790
<i>Raleigh-Cary, NC</i>	0.61246	0.40171	27	1025563	0.25178
<i>Lancaster, PA</i>	0.61010	0.38914	19	158072	0.16421
<i>Canton-Massillon, OH</i>	0.59744	0.32469	23	179319	0.21652
<i>Charleston-North Charleston, SC</i>	0.59041	0.36747	27	536832	0.17679
<i>Fresno, CA</i>	0.58439	0.44735	17	939322	0.36258
<i>Birmingham, AL</i>	0.58380	0.26010	88	839791	0.08090
<i>Denver-Aurora-Centennial, CO</i>	0.57209	0.41126	45	2326229	0.14236
<i>Austin-Round Rock-San Marcos, TX</i>	0.56872	0.44954	49	1708042	0.33462
Bottom 10					
<i>Minneapolis-St. Paul-Bloomington, MN-WI</i>	0.22379	0.75992	212	3400027	0.03282
<i>Detroit-Warren-Dearborn, MI</i>	0.24197	0.71880	109	2853011	0.06545
<i>Akron, OH</i>	0.24989	0.58605	31	543087	0.15050
<i>Waterbury-Shelton, CT</i>	0.27871	0.64736	6	278468	0.25592
<i>Bridgeport-Stamford-Danbury, CT</i>	0.29664	0.70978	5	464534	0.26150
<i>Buffalo-Cheektowaga, NY</i>	0.29898	0.83347	26	503227	0.32275
<i>Worcester, MA</i>	0.29936	0.72778	5	330111	0.42952
<i>Jacksonville, FL</i>	0.30860	0.69802	16	1067907	0.80036
<i>Milwaukee-Waukesha, WI</i>	0.31496	0.50255	63	1446632	0.16765
<i>Springfield, MA</i>	0.31588	0.75612	7	357978	0.24874

I report a summary table for rectangular bounding calculations in Table 2. A similar table for circular bounds can be found in Appendix A Table 4. Calculations are performed for the top 100 metropolitan statistical areas by population, with the highest and lowest 10 SDI scores extracted from this sample. 5 of the highest 10 scores (most spatially irregular) are found in the South, whereas 5 of the lowest are in the Midwest, potentially reflecting regional

disparities. The SDI appears loosely correlated with the HHI, though this is not a definite pattern: the Minneapolis-St. Paul-Bloomington MSA (the least *spatially* fragmented region) possesses a far lower HHI than that of the Wichita MSA (the most spatially fragmented). Lastly, I note the low municipal figures for MSAs in the Northeast, which is likely due to differences in municipal categorization and may be skewing results. In the empirical results, I introduce state-level fixed effects to control for such trends, though they are important to note when interpreting the results of Table 2.

3.3 Firm location

The most common dataset used to assess firm location in the US is the Census’s County Business Patterns (CBP). However, while CBP provides detailed information of firm industry and employment, its most granular aggregation is at the ZIP code level, which complicates an investigation of agglomeration economies that may be most relevant at scales of miles or hundreds of feet³, beyond which spatial spillovers are far less meaningful. Given an average zip code radius of roughly two miles, this fact poses a significant challenge to analyzing the potential welfare effects of distribution across space, as the patterns of firm location *within* ZIP codes may be even more relevant than those *across* ZIP codes.

To address this, I use a “matching” procedure⁴ to generate probabilistic firm locations based upon census block-level employment characteristics. This data is found in the Longitudinal Employer-Household Dynamics (LEHD) Workplace Area Characteristics (WAC) dataset from the US Census. This data has the advantage of providing information at the finest possible spatial scale, noted in Census documentation as the “smallest level of geography” at which one can procure basic demographic data (U.S. Census Bureau, 2011). Using block-level unique geographic identifiers present in the data, I then match LEHD rows to TIGER polygon centroids to create an enriched set of points that contain each block’s

³A relevant example is Arzaghi and Henderson (2008), who finds attenuation of proximity benefits happens at a scale of blocks for advertising firms in New York City.

⁴A similar approach is used in Buzard et al. (2017)

respective WAC data.

Following this, I use HUD ZIP-CBSA crosswalk files to procedurally match each census block to a ZIP Code Tabulation Area (ZCTA). Because these can occasionally cross county and CBSA lines, I adjust ZCTA employment by the percentage within the needed CBSA: an example of total CBSA level establishments by ZCTA is available for the Seattle MSA in Appendix B Figure 5. To determine the locations of firms, each block is given a weight as determined by their share of employment within the ZCTA code. This matching procedure can be refined to locate firms given the first two characters of their NAICS code, which is given in WAC employment data. The final output is a distinct geographic point for every firm in a given metropolitan area enriched with predicted firm size and industry. A sample of points within the Seattle city limits is given in Appendix B Figure 6.

3.4 Firm Concentration

To transform an array of points into a global measure of dispersion/concentration, I turn to the Duranton-Overman Kd function. I provide a brief overview of the technical assumptions behind both of these measures, and perform a sample calculation for the previously constructed establishment-point data in the Seattle-Tacoma-Bellevue MSA.

Duranton and Overman (2005) give their primary estimator of localization for n establishments as

$$K(d) = \frac{1}{n(n-1)h} \sum_{i=1}^{n-1} \sum_{j=i+1}^n f\left(\frac{d - d_{ij}}{h}\right)$$

where d is the distance at which the K -density is estimated, d_{ij} is the Euclidean distance between establishments i and j , f is a Gaussian kernel with bandwidth h . They also construct a *weighted* estimator with $e(i)$ giving a weight (e.g. employment) of firm i as

$$K^e(d) = \frac{1}{h \sum_{i=1}^{n-1} \sum_{j=i+1}^n e(i)e(j)} \sum_{i=1}^{n-1} \sum_{j=i+1}^n e(i)e(j) f\left(\frac{d - d_{ij}}{h}\right)$$

using the same Gaussian kernel and bandwidth as above. Their choice of a probability

density function as opposed to a cumulative function like Ripley’s L or the Marcon and Puech (2010) M function is able to more accurately capture densities at finer spatial scales⁵. Notably, the K -density function compares industry clustering against a counterfactual of manufacturing, as opposed to complete spatial randomness, which controls for the level of clustering present in cities that can be attributed to the distribution of a finite set of n points⁶.

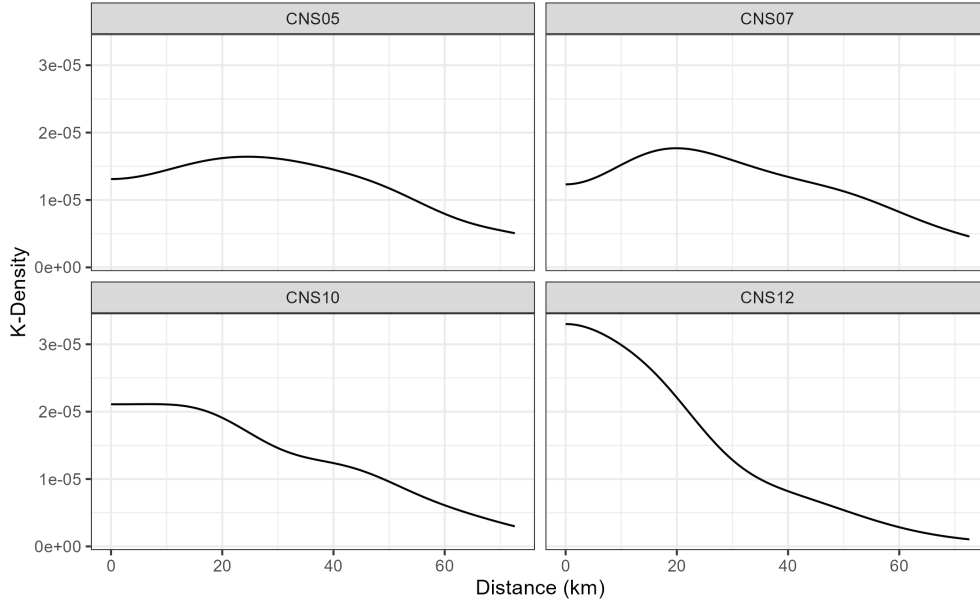
To provide basic intuition about the shape of the results, I estimate the Kd function for 4 distinct sets of points from the LEHD WAC data. These fall into the bins *CNS05*, *CNS07*, *CNS10*, and *CNS12*, corresponding with two-digit NAICS codes of 31-33 (Manufacturing), 44-45 (Retail Trade), 52 (Finance and Insurance), and 54 (Professional, Scientific, and Technical Services) respectively. From Duranton and Overman (2005), one expects manufacturing to score the lowest, whereas industries requiring higher degrees of face-to-face interaction score higher. For illustrative purposes, I take a computationally simpler approach and treat each LODES block as a singular point, with the number of employees in that block serving as weights.

The results of this preliminary investigation are reported in Figure 1. The theoretical expectations are confirmed: at distances below 40 kilometers, manufacturing displays the lowest degree of clustering, with NAICS 52 establishments displaying the highest. I note that at higher distances, this effect reverses, with the “clustering” of manufacturing persisting for higher distances than specialized NAICS codes. This is an expected result: if many points are clustered at smaller spatial scales, casting a wider net will slowly reduce the density score. In contrast, more dispersed patterns will yield increasing (or stable) scores as the distance increases. From these basic results showing divergence between industries at the Census block-employment level, I take the Duranton and Overman (2005) Kd measure to

⁵For instance, a Kd estimate would be able to give the exact density at distance d , whereas the Ripley L estimate calculates all points within distance d , which would not allow a derivation of the exact measured distance at which clustering ceases.

⁶Duranton and Overman (2005) provide the example of the vacuum cleaner industry, in which a large percentage of the workforce is concentrated into a small number of plants, which can produce spatial unevenness without any meaningful agglomeration

Figure 1: K-Density Calculations for LODES Industries



be an appropriate indicator of industrial agglomeration.

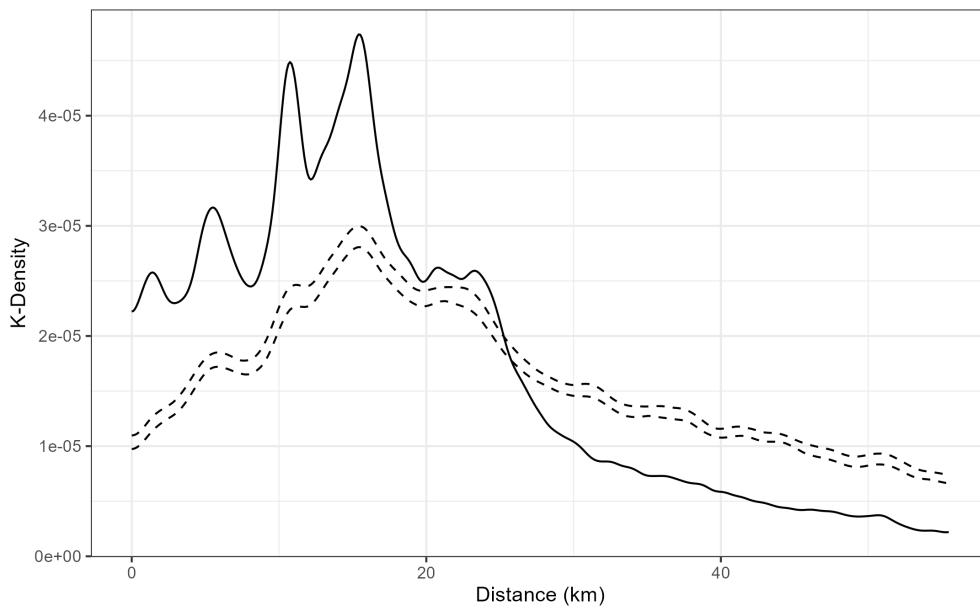
3.5 High-tech Clusters

The primary collection of establishment NAICS codes reviewed in the empirical section will be those belonging to the tech industry. I select these for two reasons. The first is that the tech industry has been shown to be prone to clustering (Kerr & Robert-Nicoud, 2020), in large part because of the critical role that knowledge spillovers play in the innovative process (Atkin et al., 2022). Additionally, tech establishments themselves tend to be attractive to municipalities seeking economic advantage, both at the national scale as demonstrated by the competition for Amazon’s HQ2, but also within metropolitan areas (Muller, 1997).

I select the relevant NAICS codes using those identified by Goldschlag and Miranda (2020) as belonging to the “high-tech” industry, and further narrow these to codes within the top 20 nationally by establishment entry as measured in Decker and Haltiwanger (2024). This yields 4 NAICS codes: 5415 (*Computer Systems Design and Related Services*), 5112 (*Software Publishers*), 5182 (*Data Processing, Hosting, and Related Services*), and 5417

(*Scientific Research and Development Services*). I perform the procedure outlined in Section 3.3 to identify block-level establishment probabilities. I then round these estimates, selecting an additive parameter that yields the same total establishments as those specified in the *ZBP* dataset to get whole number establishment-points. Finally, I perform a spatial “jitter” to distribute these points within a radius of the original block centroid. Given a median block radius of approximately 50 meters, I set this as the maximum distance that a point can fall from the center. To construct the global bandwidth as in Duranton and Overman (2005), I calculate a similar set of points for the manufacturing industry (NAICS 31-33) to serve as a baseline.

Figure 2: K-Density Calculation for Seattle MSA High-Tech Establishments



Estimates are shown in Figure 2. The solid line shows the K -density estimate for high-tech establishments, whereas the two dashed lines give the global confidence interval. At all distances where the estimated Kd lines above this interval, agglomeration can be observed: this is true of all distances less than 20 kilometers.

4 Results

4.1 Small Streams

An ideal OLS estimate of municipal fragmentation’s effect upon agglomeration would be written as

$$Agg_{jt} = \alpha + \beta Frag + \gamma X_j + \delta_j + \lambda_t + \epsilon_{jt}$$

Where Agg is a global measure of spatial agglomeration for CBSA j in year t , X is a vector of time-variant metropolitan area-level controls, and δ and λ are CBSA and year fixed effects, respectively. The parameter of interest is β , which captures the “fragmentation” of a given CBSA as measured through the indices described above. However, this approach presents a number of concerns. First of which is the considerable risk of omitted variable bias: the degree of municipal fragmentation may be determined by a range of factors that directly affect how firms choose to locate across space. For instance, rough terrain may have influenced initial settlement patterns, but also continues to affect where and how firms cluster. Moreover, there is a large reverse causality issue. Firms may locate in response to municipal incentives, but the creation of new municipalities may also follow patterns of economic activity. Thus, separating out the independent effect of municipal fragmentation on spatial agglomeration is crucial.

To do this, I instrument fragmentation with the length of small streams within a metropolitan area. Adopted from Hoxby (2000) and later used to directly measure municipal fragmentation in Hatfield et al. (2022), this approach identifies small, easily traversable waterways as natural borders at which to establish political boundaries that do not have a significant effect on the location of economic activity. Indeed, I argue that given the irrelevance of such waterways in any meaningful way for a firm’s production activities, they would not act as a dispersing or concentrating force at the metropolitan area aside from their effect on municipal boundaries. In this specification, the first stage equation can be written as such:

$$Frag = \delta_0 + \delta_1 Streams + \Theta X_j + \delta_s + v_{jt}$$

Where s are state fixed effects and δ_1 is the parameter of interest. X is a set of topographic controls at the CBSA level, as it may be the case that while small streams do not directly affect firm location decisions, they can be correlated with other geographical characteristics (larger bodies of water, hilly terrain, etc.) that do exert a larger influence on agglomeration.

Stream lengths were calculated using the Environmental Systems Research Institute's (ESRI) North American Waterways GIS shapefiles as matched to the boundaries of CBSAs. Dummies were generated by location on either the Great Lakes or the Atlantic and Pacific Oceans, as these large water bodies may both influence the presence of small waterways within a given CBSA as well as the distribution of jurisdictional boundaries. Terrain ruggedness was introduced as a means of controlling for the potential presence of hilly or mountainous terrain, which both potentially correlates with the existence of waterways in a region while also introducing a spatial barrier to jurisdictional consolidation. This variable is calculated from terrain ruggedness estimates at the census tract level from the USDA Economic Research Service, then aggregated as standard deviation at the CBSA level.

Results are shown in Table 2. The number of municipalities is included as an outcome to serve as a robustness check on the possibility that streams are simply serving to redistribute populations across boundaries rather than affecting the boundaries themselves. The coefficients from estimates (2) and (4) are both highly significant at the 1% level, demonstrating high confidence in the relevance of stream length as an instrument. The magnitudes of the estimates are somewhat difficult to interpret given the differences in orders of magnitude between the outcomes and independent variables, though they indicate that an increase in the length of small streams within a CBSA is associated with a greater number of municipalities and a lower HHI (greater fragmentation).

Table 3: First Stage Results for Small Streams IV

	Number of Municipalities (1)	(2)	Population HHI (3)	(4)
Stream Length (meters)	0.0000138*** (0.00000117)	0.0000104*** (0.00000118)	-5.08e-08*** (1.08e-08)	-5.41e-08*** (1.56e-08)
On Great Lakes		4.503 (2.943)		-0.0305 (0.0390)
On Ocean		-3.492 (2.003)		-0.107*** (0.0261)
Terrain Ruggedness		-0.0459 (0.0693)		0.000501 (0.000915)
Land Area (meters)		1.61e-11 (1.56e-10)		-7.49e-12*** (1.98e-12)
Total Population		0.0000246*** (0.000000718)		
Constant	7.914*** (1.054)	4.787*** (1.065)	0.528*** (0.00966)	0.577*** (0.0141)
State FE	No	Yes	No	Yes
R-squared	0.131	0.712	0.0238	0.322
Observations	916	913	916	913

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

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Appendix

A Tables

Table 4: Top and Bottom 10 MSAs by Spatial Fragmentation (Bounding Circle)

CBSA Title	SDI	SDI - Central	Num. Munis	Muni Pop.	HHI - Muni Pop.
Top 10					
<i>Wichita, KS</i>	0.78941	0.16862	50	577787	0.47521
<i>Bakersfield-Delano, CA</i>	0.73687	0.26980	11	608819	0.46392
<i>Tulsa, OK</i>	0.73337	0.38797	73	807991	0.28790
<i>Charleston-North Charleston, SC</i>	0.72727	0.27721	27	536832	0.17679
<i>Tampa-St. Petersburg-Clearwater, FL</i>	0.71489	0.35734	35	1194072	0.17252
<i>McAllen-Edinburg-Mission, TX</i>	0.70541	0.30349	22	625112	0.12761
<i>Lancaster, PA</i>	0.70478	0.29434	19	158072	0.16421
<i>Denver-Aurora-Centennial, CO</i>	0.70315	0.26232	45	2326229	0.14236
<i>Phoenix-Mesa-Chandler, AZ</i>	0.68616	0.36525	33	4406705	0.17058
<i>Canton-Massillon, OH</i>	0.68563	0.24752	23	179319	0.21652
Bottom 10					
<i>Worcester, MA</i>	0.41927	0.58543	5	330111	0.42952
<i>Jacksonville, FL</i>	0.43780	0.57347	16	1067907	0.80036
<i>Waterbury-Shelton, CT</i>	0.43908	0.55582	6	278468	0.25592
<i>Akron, OH</i>	0.47203	0.47853	31	543087	0.15050
<i>Minneapolis-St. Paul-Bloomington, MN-WI</i>	0.49068	0.47014	212	3400027	0.03282
<i>Virginia Beach-Chesapeake-Norfolk, VA-NC</i>	0.49335	0.60532	15	1501107	0.17762
<i>New Haven, CT</i>	0.49398	0.42905	5	303282	0.29924
<i>Detroit-Warren-Dearborn, MI</i>	0.49494	0.49424	109	2853011	0.06545
<i>Springfield, MA</i>	0.49962	0.62826	7	357978	0.24874
<i>Bridgeport-Stamford-Danbury, CT</i>	0.50005	0.48440	5	464534	0.26150

B Figures

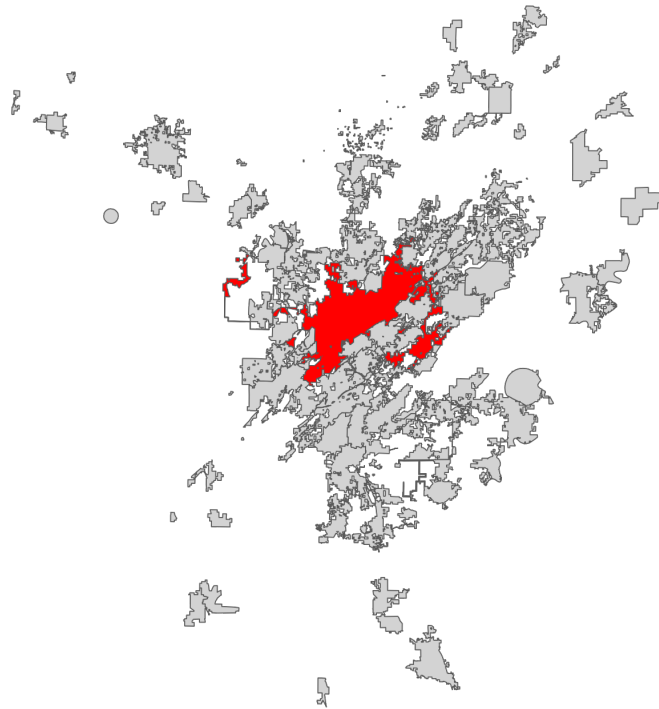


Figure 3: Municipalities in the Birmingham, AL Metropolitan Statistical Area. Birmingham City highlighted in red

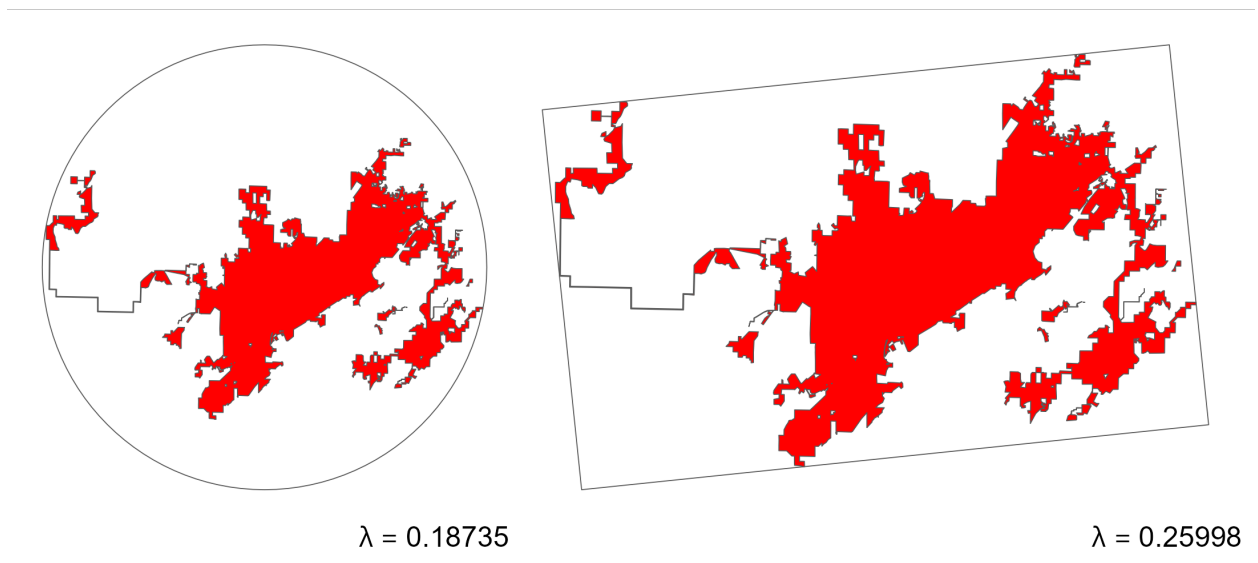


Figure 4: Circular and Rectangular Buffers for Birmingham City

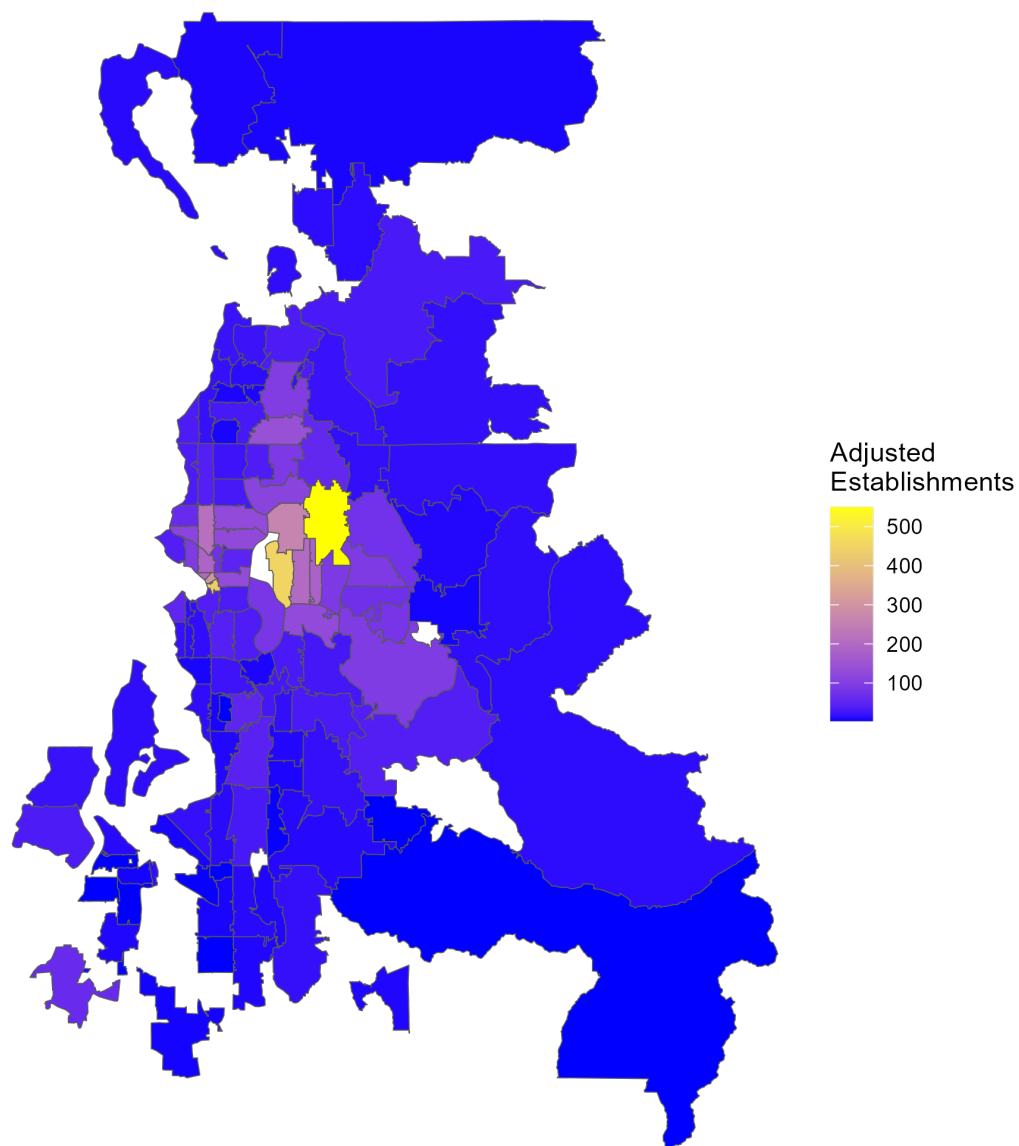


Figure 5: ZBP Adjusted High Tech Establishments for the Seattle-Tacoma-Bellevue MSA

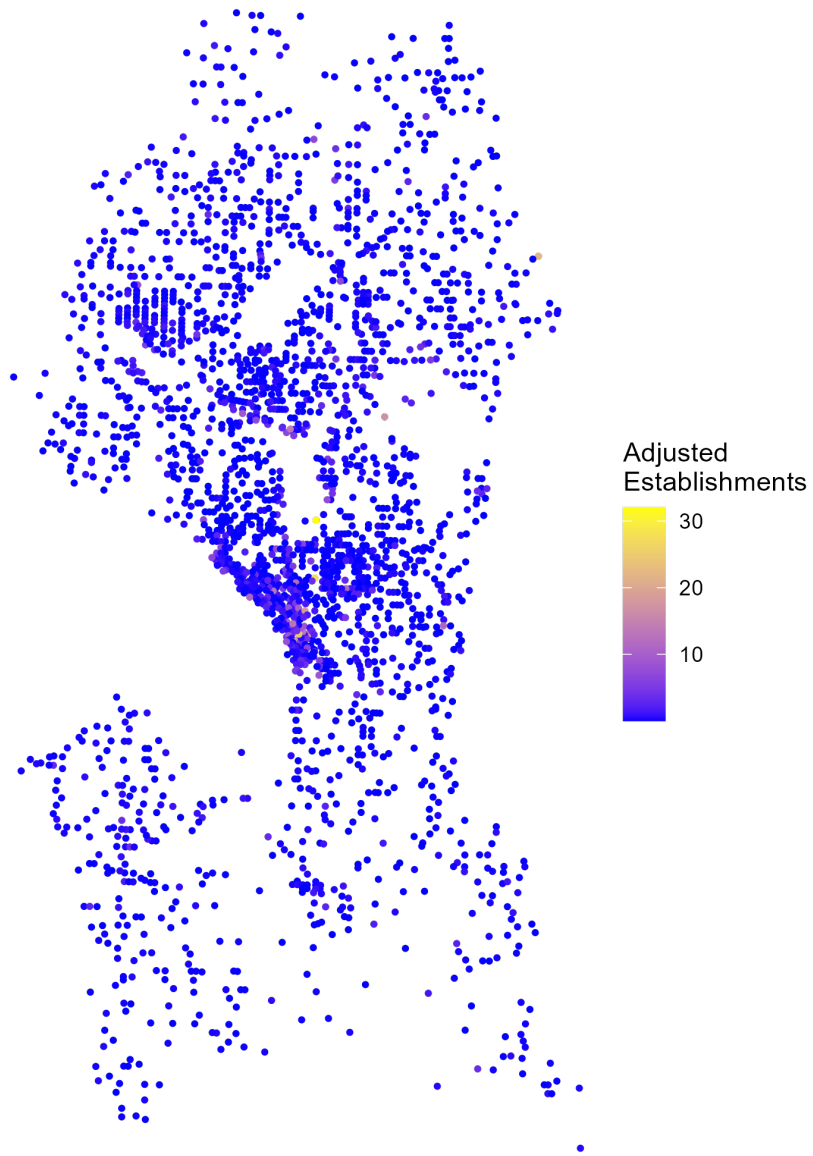


Figure 6: High Tech Establishment Matched Points within the Seattle City Limits